

Announcements

- ▶ Any questions/issues?

- ▶ Schedule:

Today (Follow-up): Discussion Q2 — T_{Earth}

Today (New): Milankovitch cycles — long-time perturbations of temperature

Tomorrow (Lab): clouds!

Thursday (Discussion): Paleotemperatures and Milankovitch

Next Week: “Climate sensitivity” — CO_2 effect on temperature

Disc 09 — Atmosphere & Lapse Rate

The layers of the atmosphere:

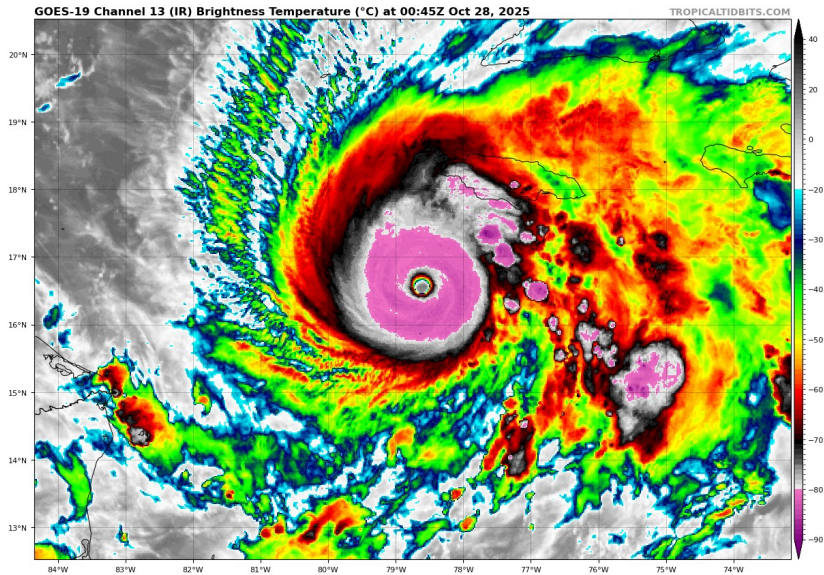
- **Troposphere** [0–10 km]: Region of weather, contains 80% of total air mass.
- **Stratosphere** [10–50 km] Ozone region where near-UV solar radiation is absorbed; temperatures increase to 0.°C
- **Mesosphere** [50–85 km] Low density of air (and ozone) causes temperature decrease to –100°C.
- **Thermosphere** [> 85 km] Ionization of molecules by far-UV solar radiation causes temperatures to reach 500–2000°C but pressure and density is extremely low (the “ionosphere” is here).

Temperature change in the troposphere is dictated by the *adiabatic lapse rate*:

$$\frac{\Delta T}{\Delta h} = -\Gamma \quad \text{w/} \quad \Gamma \approx 6.5 \frac{^{\circ}\text{C}}{\text{km}}$$

which follows from air and water vapor *convection*. The value above is the “moist” lapse rate because it includes the effect of water vapor convection; a “dry” lapse rate can be derived exactly from thermodynamics ($\Gamma_{\text{dry}} = 10 \frac{^{\circ}\text{C}}{\text{km}}$).

Hurricane Melissa



Discussion Q2

2. **Radiative-Convective Equilibrium Temperature** In prior weeks we made estimates of the Earth's equilibrium temperature. In fact this is the driving question of the first half of the course: What do we expect to be the temperature of the Earth? Let's review our prior results and then push forward to a new (and basically final) answer using the mechanism of energy transfer by evaporation.
- (a) Together with your group, remind yourselves of the two calculations we did previously for estimating the Earth's surface temperature: (i) Assuming no atmosphere, only the incoming solar flux ($S = 1360 \text{ J/s/m}^2$) and the average albedo of $\alpha = 0.3$; (ii) With an atmosphere that absorbs and then re-emits all of the Earth's outgoing (infrared) flux. What were the temperatures found in each case, and how do they compare to the observed average temperature of $\sim 15^\circ\text{C}$?
 - (b) The *lapse rate* mentioned implies an additional flux of heat from the Earth's surface to the atmosphere. Redo the energy flow (or flux) diagram (e.g., Fig 4.5 of our textbook), but now include an additional arrow for "convective flux," F_c , from the Earth to the atmosphere.
 - (c) Write down three energy balance equations ($\text{Flux}_{\text{in}} = \text{Flux}_{\text{out}}$) related to your figure from (b): (i) for the object of the whole Earth, (ii) for the atmosphere, and (iii) for the Earth's surface.
 - (d) It turns out we do not need to "solve" these equations. We can use the *adiabatic lapse rate* together with your first equation (i) to find the temperature of the Earth's surface. Assume that the height of this atmosphere "object" is where the atmospheric pressure drops to half its surface value. What surface temperature do you obtain?

(i) No-atmosphere Earth

Let's remind ourselves about our first calculation of the equilibrium temperature of the planet. Remember that the Stefan-Boltzmann Law states

$$F = \sigma T^4 \quad \text{with} \quad \sigma = 5.7 \times 10^{-8} \frac{\text{W}}{\text{m}^2 \text{K}^4}$$

and that flux means $F = \frac{\text{energy/time}}{\text{area}}$.

EQUILIB :

$$F_{\text{in}} = F_{\text{out}}$$

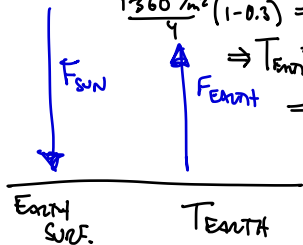
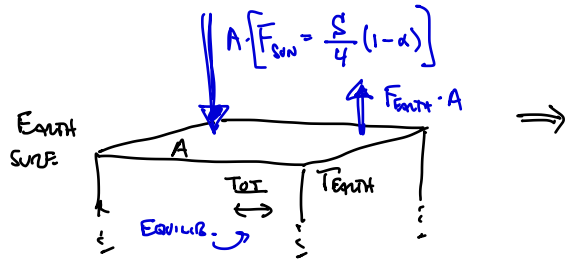
$$F_{\text{sun}} = F_{\text{earth}}$$

$$\frac{S}{4} (1 - \alpha) = \sigma T_{\text{earth}}^4$$

$$\frac{1360 \text{ W/m}^2}{4} (1 - 0.3) = \sigma T^4$$

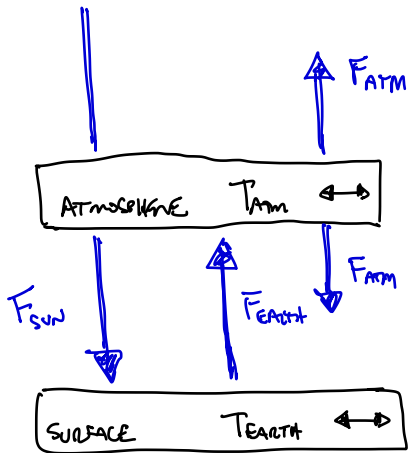
$$\Rightarrow T_{\text{earth}} = 254 \text{ K}$$

$$= -39^\circ \text{C}$$



(ii) Earth with atmosphere

Now consider the case where the Earth's surface is covered by an atmosphere of ideal greenhouse gases.



$$F_{\text{in}} = F_{\text{OUT}} \quad (\text{ATM})$$

$$F_{\text{EARTH}} = 2 F_{\text{ATM}} \quad (1)$$

$$(1) \quad F_{\text{ATM}} = F_{\text{EARTH}} / 2$$

$$\rightarrow (2) \quad F_{\text{SUN}} + \left(\frac{F_{\text{EARTH}}}{2} \right) = F_{\text{EARTH}}$$

$$\Rightarrow \boxed{F_{\text{SUN}} = \frac{F_{\text{EARTH}}}{2}}$$

$$F_{\text{in}} = F_{\text{OUT}} \quad (\text{SURFACE})$$

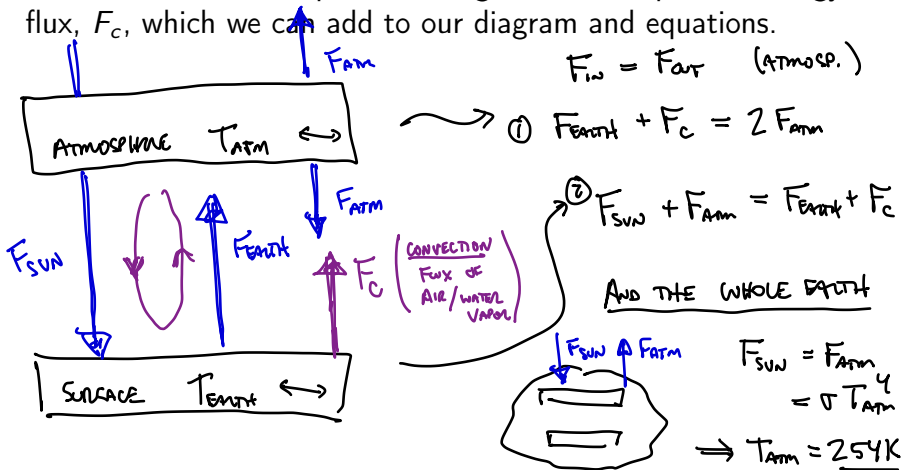
$$F_{\text{SUN}} + F_{\text{ATM}} = F_{\text{EARTH}} \quad (2)$$

2ND ESTIM.

$$T_{\text{EARTH}} = 302\text{K} = 29^{\circ}\text{C}$$

(b)–(c) Earth with a convective atmosphere

As considered in the discussion, the convection of warm air — and, most importantly, water vapor — from the Earth's surface to the upper troposphere transports energy from the surface into the atmosphere. This gives another path of energy flux, F_c , which we can add to our diagram and equations.



(d) Earth with a convective atmosphere

The solution to these equations comes simply from knowing the (moist) adiabatic lapse rate ($\Gamma = 6.5 \frac{^{\circ}\text{C}}{\text{km}}$) and the average height of the greenhouse gas layer ($h \approx 5 \text{ km}$).

$$\text{WE KNOW THAT } \frac{\Delta T}{\Delta h} = -6.5 \frac{^{\circ}\text{C}}{\text{km}}$$

SO THE GREENHOUSE ATMOSPHERE IS ABOUT :

$$\begin{aligned}\Delta T (\text{surf} \rightarrow \text{Atm}) &= \left(\frac{\Delta T}{\Delta h} \right) \cdot \underline{5 \text{ km}} = -6.5 \frac{^{\circ}\text{C}}{\text{km}} \cdot 5 \text{ km} \\ &= -32.5 ^{\circ}\text{C} \end{aligned}$$

$T_{\text{Atm}} - T_{\text{surf}} = -32.5 \text{ K}$
 $T_{\text{surf}} = T_{\text{Atm}} + 32.5 \text{ K}$

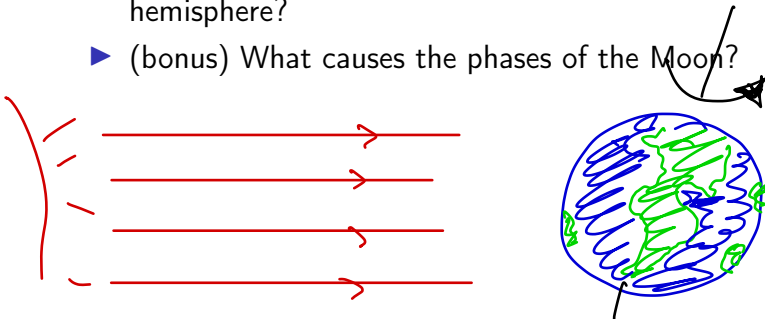
BUT $T_{\text{Atm}} = 255 \text{ K}$, so

$$\begin{aligned}\Delta T (\text{surf} \rightarrow \text{sun}) &= T_{\text{Atm}} - T_{\text{surf}} \Rightarrow T_{\text{surf}} = T_{\text{Atm}} + 32.5 \text{ K} \\ &= 287.5 \text{ K} = 14.5 ^{\circ}\text{C}\end{aligned}$$

The Earth and Sun (and Moon)

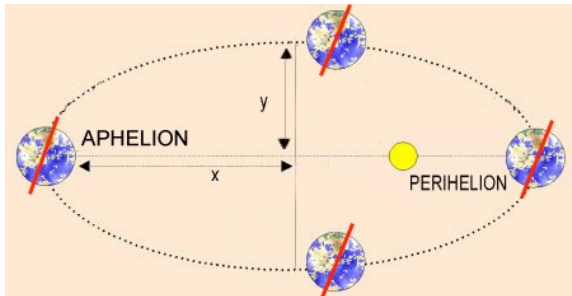
Let's start with some basic questions about the Earth's motion in space:

- ▶ What causes day and night?
- ▶ What causes the seasons in the northern (or southern) hemisphere?
- ▶ (bonus) What causes the phases of the Moon?



Seasons and eccentricity

Currently, within a hundred-thousand-year cycle, the Earth's orbit around the Sun looks like this:



(adapted from [Beth Caissie, UAF](#))

where the eccentricity (non-circular, ellipse nature of the orbit) has been exaggerated.

Where on the orbit is northern hemisphere summer?

Kepler and Newton: Two-body problem



Kepler (1571–1630)

Johannes Kepler, using planetary data collected by Tycho Brahe, found that planets and moons follow elliptical orbits.

Newton, using his dynamical law (" $F = ma$ ") and his Gravitational Force ($F_G = \frac{GMm}{r^2}$) found the theoretical origin of this observation.

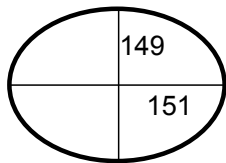


Newton (1643–1727)

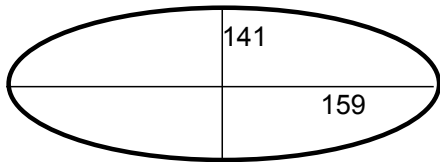
But when two orbiting bodies are “perturbed” by a third, their motion is not perfectly elliptical (the “three-body problem”).

Orbital effects on solar radiation

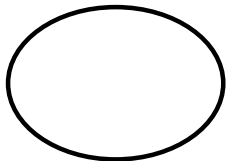
The variation of eccentricity is the the first of three orbital effects on solar radiation.



Present day

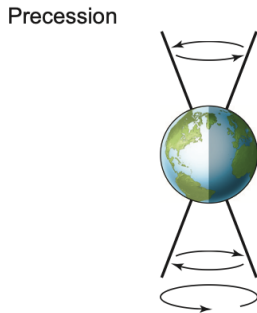
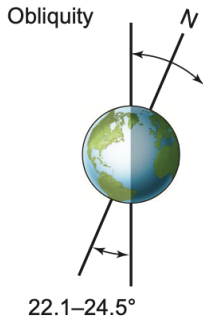
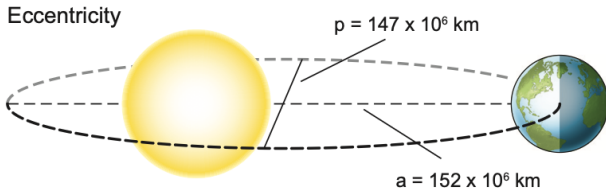


50 kya



100 kya

Orbital effects on solar radiation



Orbital effects on solar radiation

